

ATHARVA NAGANE

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EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering; CGPA: 9.16/10

Bhopal, MP

Sept 2023 – May 2027 (Expected)

EXPERIENCE

Google Summer of Code 2026 — Open Source Contributor

May 2026 – Present

RustCloud — Multi-Cloud Infrastructure Library (under C2SI)

Merged PRs

- Contributed the library's first GenAI provider layer: a unified **LlmProvider** trait spanning generation, streaming, embeddings and tool use across AWS Bedrock and GCP Vertex AI, backed by 86 unit tests.
- Designed an async OAuth2 **TokenProvider** with proactive token caching, adopted as the auth pattern for every later GCP module including BigQuery (datasets, streaming inserts, query polling).
- Mapped SDK error variants, rate limits and retryable versus terminal statuses into one **CloudError** taxonomy used by every provider; documented the design in a public blog series.

Full Stack Developer Intern

Remote

Imarticus Learning

May 2025 – Jun 2025

- Sustained **10,000+ requests/min** at a sub-1% error rate in k6 load tests after hardening Express and MongoDB APIs with rate limiting and request validation.
- Shipped an internal LMS on MERN with role-based access control, JWT auth and real-time WebSocket dashboards across 12+ course sections.
- Automated CI on GitHub Actions with parallel test runs and Vercel deploys, cutting build-to-deploy time by 25% and catching 15+ defects per sprint.

PROJECTS

Blastwave — Reliability Platform for Incident Investigation & Fault Injection

Jul 2026 – Present

Rust, Docker, Prometheus, Loki

GitHub

- Developed a fault-injection engine covering network latency and loss (tc/netem), cgroup pressure, container kills and dependency outages; **all 6 fault kinds** reproduce their documented signature.
- Built a verify-then-conclude agent over 6 read-only observability tools that refuses any conclusion unsupported by two evidence items; its deterministic reasoner identifies 5 of 6 fault kinds under blind injection.
- Implemented sandboxed drills with blast-radius containment enforced in code, dropped capabilities and per-drill networks, scoring diagnoses 0–100 and rescheduling weak skills via SM-2 spaced repetition.

Reckon — Payment Reconciliation & Ledger-Integrity Engine

Jun 2026 – Jul 2026

TypeScript, PostgreSQL, Redis, BullMQ

GitHub

- Modeled money as immutable, append-only double-entry postings in integer minor units, with balances derived from the log and a zero-sum invariant enforced by database trigger.
- Achieved **exactly-once, crash-safe** webhook processing via same-transaction idempotency and the outbox pattern; 5x replay under duplicate, out-of-order and dropped delivery yields one ledger entry.
- Detected **all 6 payment-drift classes** under adversarial event delivery, auto-resolving 5 through reversing entries while amount mismatches escalate for human review by design.
- Encoded the payment lifecycle as an explicit state machine over an append-only event log, with hold accounting for inflight authorizations; 42 tests including property-based checks.

CodeFront — Competitive Programming Platform with Sandboxed Execution

Apr 2026 – May 2026

Node.js, React, MongoDB, Redis, Docker

GitHub

- Engineered Docker-sandboxed execution for Python, C++ and Java (network-isolated, 256MB/0.5-CPU limited, pid-capped), hardened against fork bombs and escape attempts; 30 integration tests.
- Decoupled execution from HTTP behind a Bull/Redis job queue, returning submission IDs in **under 100ms** while six verdict types resolve asynchronously at configurable concurrency.
- Deployed on AWS EC2 behind nginx with SSL, a Vercel frontend and MongoDB Atlas; added Gemini-based review on accepted submissions.

TECHNICAL SKILLS

Languages: C++(proficient), Python, JavaScript, Rust, SQL

Backend & Data: Node.js, Express, REST API, authentication (JWT, OAuth2), PostgreSQL, Redis, MongoDB, BullMQ

Cloud & Tooling: Docker, AWS (EC2, Bedrock), GCP (Vertex AI, BigQuery), Prometheus, Loki, Git, CI/CD

Foundations: Data Structures & Algorithms, Operating Systems, DBMS, OOP, distributed systems patterns

ACHIEVEMENTS & CERTIFICATIONS

- **Oracle AI Vector Search Certified Professional (OCP).**
- Solved **200+** DSA problems on LeetCode; active contest participant, rating **1560**.
- **Smart India Hackathon 2024:** qualified for the national rounds; top-50 team in university internal rounds.
- Second Dan Black Belt & Sensei (WKF-affiliated), instructing 50–70 students; National Gold and International Bronze medalist.